

Vulnerability Assessment Report

1st January 20XX

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 20XX to August 20XX. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

The database server plays an important role in the business because it stores and manages valuable company and customer information. This data is used every day to support business operations, making it essential that it remains secure and accessible. If an attacker gained access to the server, sensitive information could be exposed, resulting in financial losses and damage to the company's reputation. Likewise, if the server became unavailable, employees could lose access to critical data needed to perform their jobs. Conducting a security analysis helps identify potential risks and ensures that proper protections are in place to keep the system secure.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Hacker	Can Obtain Highly sensitive information through exfiltration	3	3	9
Employee	Can disrupt operations	2	3	6
Customer	Can delete or alter information	1	3	3

Approach

When evaluating the risks to the database server, I focused on how different users or attackers could impact the system and the data it contains. Hackers were identified as the highest risk because they often target databases to steal sensitive information, which could have serious consequences for the organization. Employees were considered a moderate risk since mistakes or misuse of access privileges can accidentally disrupt operations or compromise data. Customers were assigned the lowest likelihood because they typically have limited access to internal systems, but the potential impact would still be significant if unauthorized access occurred. By examining both the likelihood and potential impact of each threat, it was possible to determine which risks require the most attention.

Remediation Strategy

To reduce the risks identified during the assessment, the organization should strengthen its access control measures and monitor user activity more closely. Implementing strong password requirements and multi-factor authentication can help prevent unauthorized access to the database. Role-based permissions should also be used so that users only have access to the information necessary for their responsibilities. In addition, all data transmitted across the network should be protected using modern encryption protocols such as TLS. Regular monitoring, auditing, and security reviews will help identify potential threats early and ensure the database remains secure over time.